15.09.26 Практика 2

Модиф. алг. Евклида

Вычисление обратного по модулю степени двойки

$\exists a^{-1} \bmod 2^b \iff a \notin (2\mathbb{N})$

$a \in 2, 2^b - 2$

$$\begin{cases} 2^b = 0 \cdot a + 1 \cdot 2^b \\ 0 = 2^b \cdot a + (-a) \cdot 2^b \\ 0 = (-2^b) \cdot a + a \cdot 2^b \end{cases} \quad \begin{pmatrix} 2^b & 0 & 1 \\ 0 & 2^b & -a \\ 0 & -2^b & a \end{pmatrix} \sim (1 \mid 2\ \beta)$$

$1 = 2 \cdot a + \beta \cdot 2^b$

$1 \overset{2^b}{\equiv} 2a \Rightarrow a^{-1} \overset{2^b}{\equiv} 2$

Задача 1: $113^{-1} \bmod 512$

$512 = 2^9$; $b = 9$, $a = 113$

$$\begin{cases} 2^9 = 0 \cdot 113 + 1 \cdot 2^9 \\ 0 = 2^9 \cdot 113 + (-113) \cdot 2^9 \\ 0 = (-2^9) \cdot 113 + 113 \cdot 2^9 \end{cases} \quad \left(\begin{array}{c|cc} 2^9 & 0 & 1 \\ 0 & 2^9 & -113 \\ 0 & -2^9 & 113 \end{array}\right)$$

$(2^9 \mid 0\ 1) \sim (2^9 \mid 2^9\ -112) \overset{M_1(\frac{1}{2})}{\sim} (2^8 \mid 2^8 - 56) \overset{M_1(\frac{1}{2})}{\sim}$

$(2^7 \mid 2^7 - 28) \overset{M_1(\frac{1}{2})}{\sim} (2^6 \mid 2^6 - 14) \overset{M_1(\frac{1}{2})}{\sim} (2^5 \mid 2^5 - 7) \overset{M_{13}(-1)}{\sim}$

$(2^5 \mid -480\ 106) \overset{M_1(\frac{1}{2})}{\sim} (2^4 \mid -240\ 53) \sim (2^4 \mid 272\ -60) \sim$

$\sim (2^3 \mid 136\ -30) \sim (2^2 \mid 68\ -15) \overset{M_{13}(1)}{\sim} (2^2 \mid -444\ 98) \sim (2^1 \mid -222\ 49)$

$M_{12}(1) \sim (2' \mid 290 \; -64) \sim (1 \mid 145 \; -32)$

Ответ: $113^{-1}_{572} \equiv 145$

$aX \overset{c}{\equiv} b$, $a \perp c$, $c \notin (2\mathbb{N})$

$x \overset{c}{\equiv} \frac{b}{a}$

$a_1 \le a$

$$\begin{pmatrix} c & | & 0 \\ a & | & 1 \\ 0 & | & 0 \end{pmatrix} \sim \dots \sim \begin{pmatrix} a_1 & | & \alpha_1 \\ \dots & | & \dots \\ a_1 & | & \alpha_1 \end{pmatrix} \quad a - a_1 \sim \dots$$ чтобы приблизить к b

Задача 2: Вычислить $\frac{153}{179} \bmod 283$

Решение:

$$\begin{pmatrix} 283 & | & 0 \\ 179 & | & 1 \\ 0 & | & 0 \end{pmatrix} M_{12}(-1) \sim \begin{pmatrix} 104 & | & -1 \\ 179 & | & 1 \\ 0 & | & 0 \end{pmatrix} M_{31}(1) \sim \begin{pmatrix} 104 & | & -1 \\ 179 & | & 1 \\ 104 & | & -1 \end{pmatrix} =$$

≤ 153 $\quad \le 153$ $\quad \le 153 - 104 = 49$

$$= \begin{pmatrix} 104 & | & 282 \\ 179 & | & 1 \\ 104 & | & -1 \end{pmatrix} M_1\left(\tfrac{1}{2}\right) \sim \begin{pmatrix} 52 & | & 141 \\ 179 & | & 1 \\ 104 & | & -1 \end{pmatrix} M_1\left(\tfrac{1}{2}\right) \sim \begin{pmatrix} 26 & | & -71 \\ 179 & | & 1 \\ 130 & | & -72 \end{pmatrix} \sim$$

≤ 49 $\quad \le 49$ $\quad \le 23$

$$M_1\left(\tfrac{1}{2}\right) \sim \begin{pmatrix} 13 & | & 106 \\ 179 & | & 1 \\ 130 & | & -72 \end{pmatrix} M_{21}(-1) \sim M_{31}(1) \begin{pmatrix} 13 & | & 106 \\ 166 & | & 178 \\ 143 & | & 34 \end{pmatrix} \sim \begin{pmatrix} 13 & | & 106 \\ 83 & | & 89 \\ 143 & | & 34 \end{pmatrix} \sim$$

≤ 23 $\quad \le 10$ $\quad \le 10$

$M_{21}(-1)$ $\begin{pmatrix} 13 & 106 \\ 70 & 266 \\ 143 & 34 \end{pmatrix}$ $M_2(\frac{1}{2})$ $\begin{pmatrix} 13 & 106 \\ 35 & 133 \\ 143 & 34 \end{pmatrix}$ $M_{21}(1)$ $\begin{pmatrix} 13 & 106 \\ 22 & -256 \\ 143 & 34 \end{pmatrix}$

≤ 10 ≤ 10 ≤ 10

$M_2(\frac{1}{2})$ $\begin{pmatrix} 13 & 106 \\ 11 & -128 \\ 143 & 34 \end{pmatrix}$ $M_{12}(-1)$ $\begin{pmatrix} 2 & 234 \\ 11 & -128 \\ 153 & 72 \end{pmatrix}$ Ответ: 72

≤ 10 $M_{3,1}(5)$ ≤ 0

т.к. 2 меньше

10 в 5 раз

$(2d^2+d+1)(d^3+2d+2) \bmod ((d^4+d^3+2d+1), 3)$

$(2,1,1)(1,0,2,2) \bmod ((1,1,0,2,1), 3)$

$$d = \frac{\begin{vmatrix} (2,0,4,4) & (1,0,2,2) \\ (2,0,1,1) & (2,0,1,1,0) \end{vmatrix}}{(1,1,0)} \quad \frac{\begin{vmatrix} (1,0,2,2) & (1,0,2,2) \\ (2,1,2,0,0) & (2,2,2,2,2) \end{vmatrix}}{(2,1,2,0)}$$

$$- \frac{\begin{vmatrix} (1,1,0,2) & (1,0,2,2) \\ (2,1,1,0,2) & (2,2,2,2,2) \end{vmatrix}}{(2,2,0,1,2)} \quad \frac{\begin{vmatrix} (1,1,0,2) & (1,0,2,2) \\ (2,1,1,0,2) & (2,2,2,2,2) \end{vmatrix}}{(2,1,0)} = 2d^2+d$$

$P_{11}(2)$ $\ell(11) = \ell(11) = 10$

$d(10) = \{1, 2, 5, \underline{10}\}$

$2^2 = 4 \neq 1$

$2^5 = 10 \neq 1$ $2^{10} = -1 \cdot -1 = 1$

$= -1$

$\frac{4}{9} \bmod 19$

$$\left(\begin{array}{c|c} 9 & 1 \\ 19 & 0 \\ 0 & 0 \end{array}\right) \underset{\leq 4}{\sim} M_{12}(-1) \left(\begin{array}{c|c} 9 & 1 \\ 10 & 18 \\ 0 & 0 \end{array}\right) \underset{\leq 4}{\sim} M_2\left(\frac{1}{2}\right) \left(\begin{array}{c|c} 9 & 1 \\ 5 & 9 \\ 0 & 0 \end{array}\right) \underset{\leq}{\sim} M_{21}(-1)$$

$$\left(\begin{array}{c|c} 4 & 11 \\ 5 & 9 \\ 4 & \textcircled{11} \end{array}\right)$$

$$(1+2)^{-1} \bmod \left((1^2+1), 3\right)$$

$$\left(\begin{array}{c|c} (1,2) & 1 \\ (1,0,1) & 0 \end{array}\right) \underset{\sim}{M_{12}(1,0)} \left(\begin{array}{c|c} (1,2) & 1 \\ (1,1) & (2,0) \end{array}\right) \underset{\sim}{M_{21}} \left(\begin{array}{c|c} (0,1) & (1,1) \\ (1,1) & (2,0) \end{array}\right)$$

$$\left(1^2+2\cdot 1+1\right)^{-1} \bmod \left((1^3+2), 5\right)$$

$$(1,2,1) \bmod \left((1,0,0,2), 5\right)$$

$$\left(\begin{array}{c|c} 1,2,1 & 1 \\ 1,0,0,2 & 0 \end{array}\right) \underset{\sim}{M_{12}(1,0)} \left(\begin{array}{c|c} (4,2,1) & 4 \\ (3,4,2) & 4,0 \end{array}\right) \underset{\sim}{M_{12}(3)}$$

$$\begin{array}{r} 347 \\ -\ 313 \\ \hline 34 \end{array}$$

$$\begin{array}{r} 4,0 \\ -\ 3 \\ \hline 4,2 \end{array}$$

$$\left(\begin{array}{c|c} (1,2,1) & 1 \\ (3,4) & 4,2 \end{array}\right) \underset{\sim}{M_{21}(2,0)} \left(\begin{array}{c|c} 4,1 & 2,1,1 \\ 3,4 & 4,2 \end{array}\right) \underset{\sim}{M_{21}(2)}$$

$$\begin{array}{r} 1,2,1 \\ -\ 1,3,0 \\ \hline 4,1 \end{array}$$

$$\left(\begin{array}{c|c} 4,1 & 2,1,1 \\ 2 & 1,2,0 \end{array}\right) \underset{\sim}{M_2(3)} \left(\begin{array}{c|c} 4,1 & 2,1,1 \\ 1 & 3,1,0 \end{array}\right)$$

$$\begin{array}{r} 0,0,1 \\ -\ 3,4,0 \\ \hline 2,1,1 \end{array}$$

$$\begin{array}{r} 3,4 \\ -\ 3,2 \\ \hline 2 \end{array}$$

$$\begin{array}{r} 4,2 \\ -\ 4,2,2 \\ \hline 1,2,0 \end{array}$$